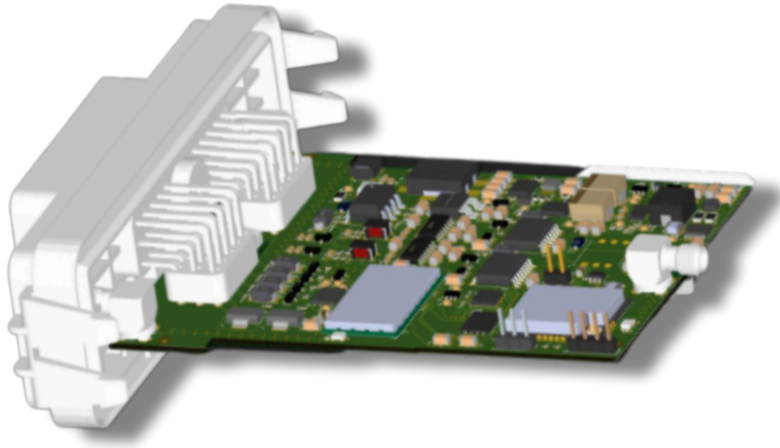


IOT CAN GATEWAY - ENEDYM (CO-OP SUMMER 2026) enedym



What?

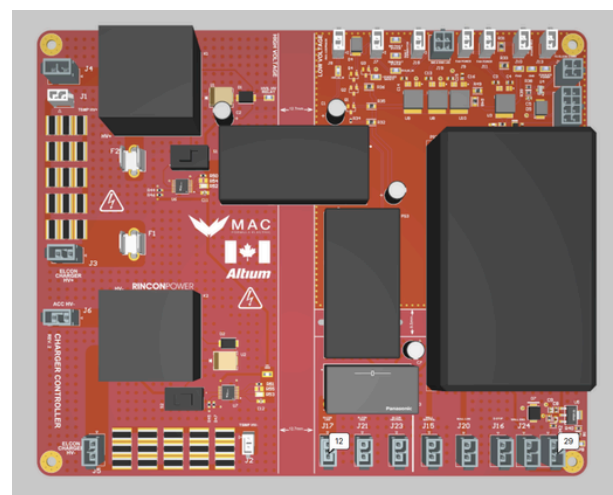
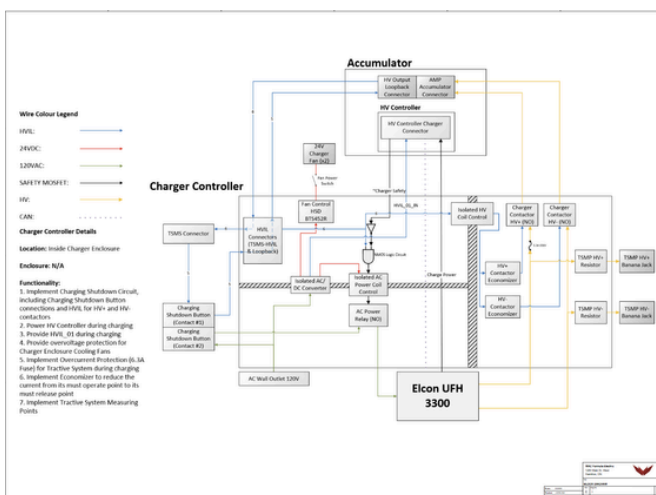
- Independently designed the complete hardware for an in-house **IoT CAN Bus Data Logger**, **cutting per-device cost by over \$500** while improving field reliability and connectivity.

How?

- Designed **6-layer PCB** in **Altium Designer**, integrating a **microcontroller**, **cellular/GNSS module**, and **CAN transceivers**.
- Developed **C firmware** for **real-time CAN bus acquisition**, **decoding**, and **wireless telemetry upload**.

Results

- **Delivered** a sealed, rugged **device** that streams live **CAN telemetry** over **cellular/Wi-Fi** and buffers data locally until reconnected.
- **Cut per-device cost by over \$500** versus the off-the-shelf unit.

CHARGER REV3 PCB - MAC FORMULA ELECTRIC 

What?

- Led charger controller redesign with **13 hardware improvements**, enhancing **safety, efficiency**, and **communication reliability**.

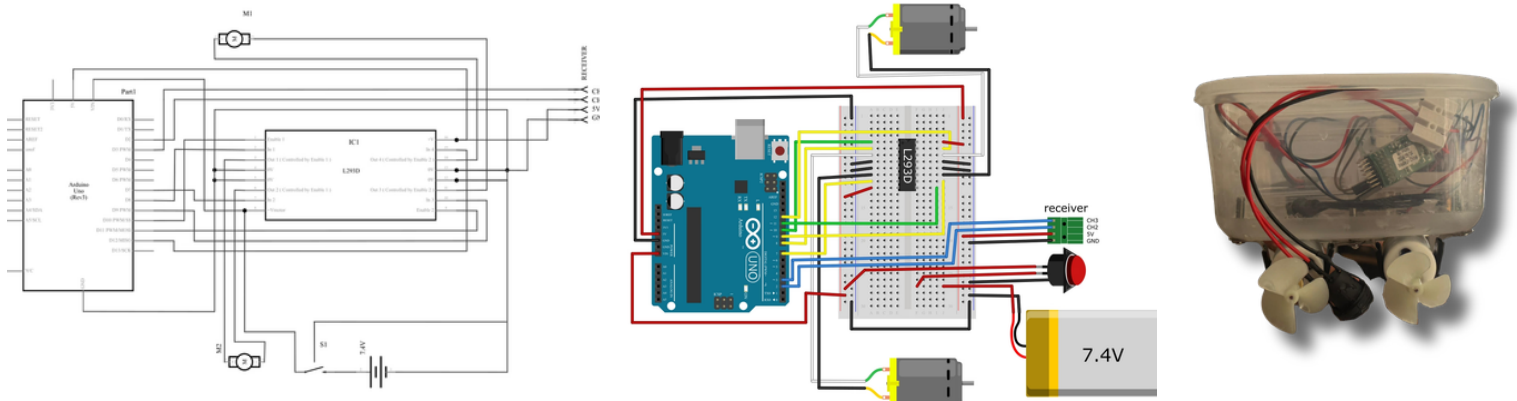
How?

- Added **NMOS safety logic**, **HV fuse**, and **240VAC-24VDC converter**, cutting components by 15%.
- **Redesigned PMOS circuit** for REV2 fix and four fail-safe states.
- **Integrated CAN bus** and **optimized PWM economizer** via PCB testing.

Results

- Achieved **stable, fail-safe charging control** for **67.2V accumulator system**.
- Improved **system safety** and **thermal performance**, validated through **impedance, smoke, and thermal** tests.

MARINE DEBRIS RC BOAT



What?

- Built a **wireless aquatic robot** for debris removal and surface cleanup.
- Used **dual brushless motors** and **LiPo power** for propulsion.

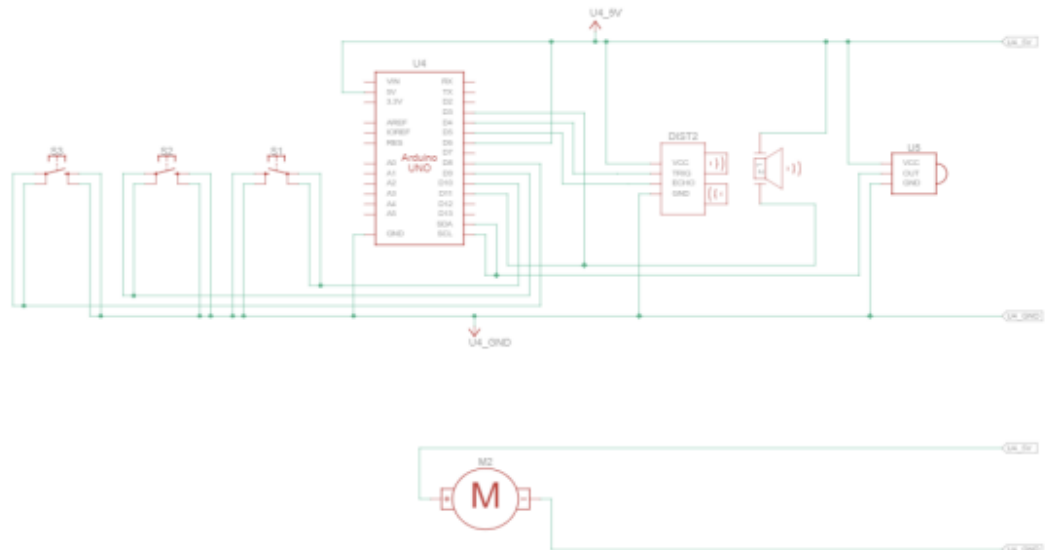
How?

- Integrated **Arduino Uno** with **RF control** and **ESC-driven motors**.
- Designed waterproof housing and efficient power circuit for stability.

Results

- Achieved smooth wireless control and **strong thrust performance**.
- Ran **40+ minutes per charge** with reliable debris collection.

SEE-R OBJECT AND TEMPERATURE DETECTION DEVICE



What?

- Built **SEE-R**, a wearable device using **IR** and **ultrasonic sensors** for heat and obstacle detection.
- Powered by an **Arduino Uno** with vibration and audio feedback.
- Designed for lightweight, hands-free operation and user comfort.

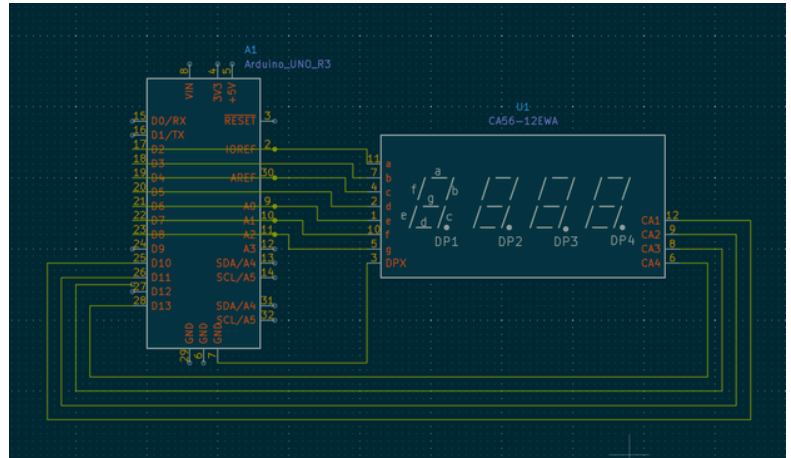
How?

- Prototyped circuits** and validated sensor integration via **TinkerCAD** and **breadboard testing**.
- Integrated all components with precise wiring, button extensions, and vibration module alignment.

Results

- Achieved **>90% detection accuracy** in real-world testing.
- Total weight under 300 g with stable sensor performance.
- Delivered a **fully functional prototype** with **reliable vibration** and **audio alerts**.

HP3435A MULTIMETER REVERSE-ENGINEERING PROJECT



What?

- Reverse-engineered a vintage **HP3435A digital multimeter** with a **failed display system**.
- Built an **Arduino-based data acquisition system** to capture **voltage and current readings** in real time.

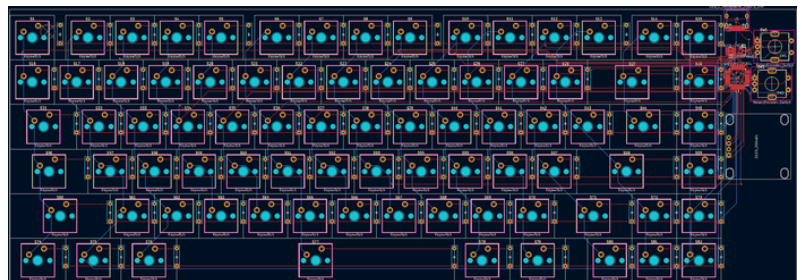
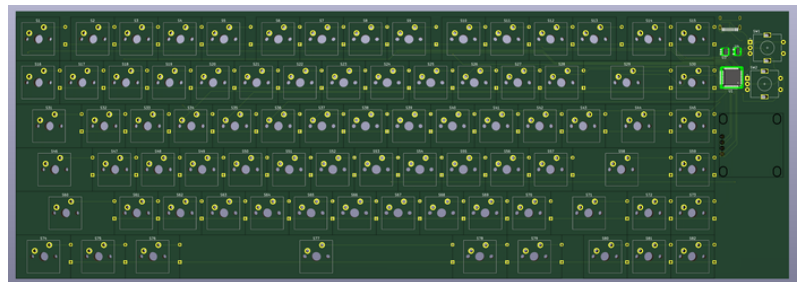
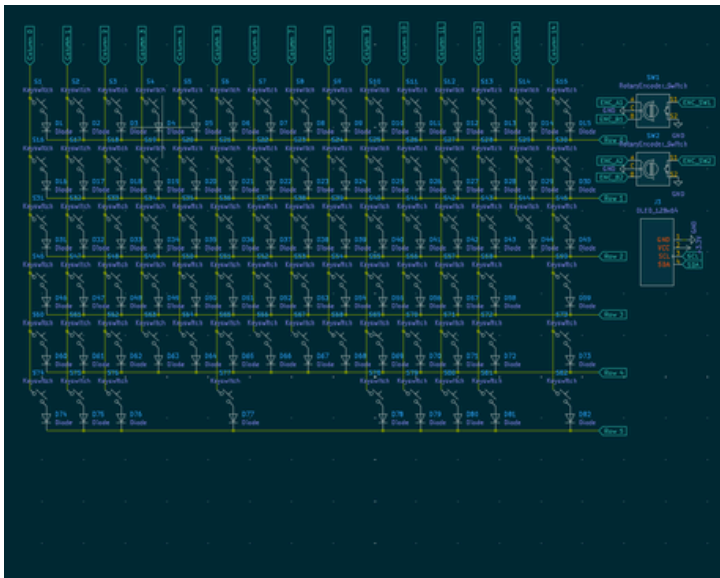
How?

- Interfaced Arduino Uno** with the **multimeter's display driver** via digital I/O mapping.
- Implemented **serial data logging to Excel** for real-time graphing and calibration comparison.

Results

- Successfully restored measurement readouts** via serial monitor and Excel dashboard.
- Validated readings** against reference equipment with **<2% deviation**.

STM32 BASED MECHANICAL KEYBOARD



What?

- Designed and built a custom **82-key Hall-effect USB keyboard**
- With **STM32 MCU**, **OLED display**, **dual rotary encoders**, and **USB-C**.

How?

- Developed **full PCB in KiCAD**, integrating power management, USB, and sensor circuits.
- Prototyped, assembled, and debugged hardware**, optimizing firmware for stable performance.

Results

- Delivered a fully functional, **low-latency keyboard** with accurate **analog key sensing**.
- Achieved **stable USB communication** and efficient power delivery under load.